

Design Task

1. Hybrid AI Recommendation Engine Design

Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of

intelligent agents in adapting recommendations based on user behavior and feedback in real

time.

2. Smart Disaster Detection System

Develop a distributed AI system using PySpark and RDD operations to detect natural disasters

(such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and

social media). Describe the system architecture, data fusion techniques, and how intelligent

agents coordinate to provide early warnings and response strategies.

3. AI-Based Fraud Detection Framework

Design a scalable fraud detection system using PySpark RDDs to process large volumes of

financial transactions. Explain how anomaly detection algorithms can be implemented in a

distributed manner and how intelligent agents collaborate to identify suspicious patterns,

trigger alerts, and continuously improve detection accuracy.

4. Autonomous Supply Chain Optimization System

Construct a distributed AI system using PySpark for optimizing supply chain operations

(inventory, logistics, demand forecasting). Describe how RDD transformations support large-

scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

5. Personalized Learning Analytics Platform

Design an AI-driven educational platform that uses PySpark and RDDs to analyze student

learning behavior and performance data. Explain how the system supports adaptive learning

through intelligent agents, real-time feedback, and scalable analytics to personalize content

delivery for thousands of learners.